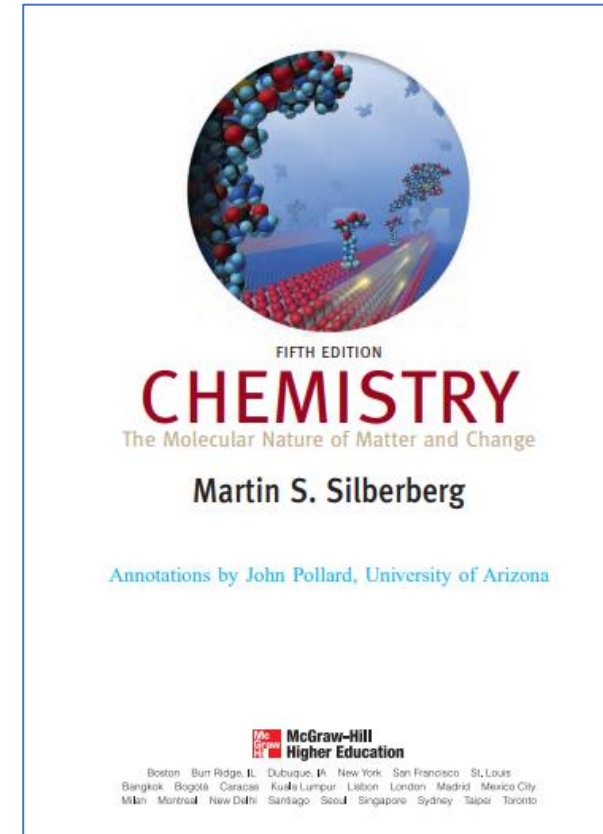
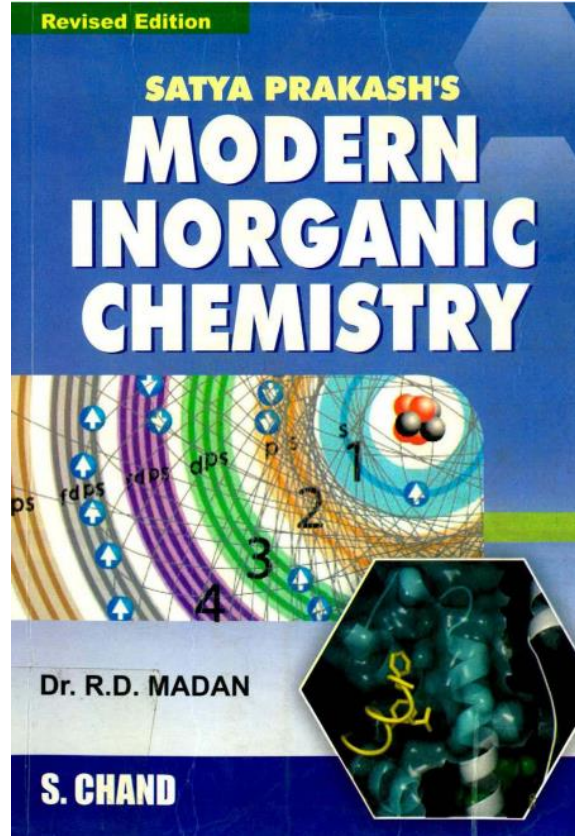


GENERAL CHEMISTRY – CHE 1201

Chemistry of Elements

p-block Halogens

Referencing Texts



Group 7A(17): The Halogens

KEY

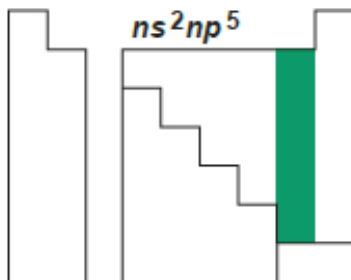
Atomic No.

Symbol

Atomic mass

Valence e⁻
con gur ation

Common
oxidation
states



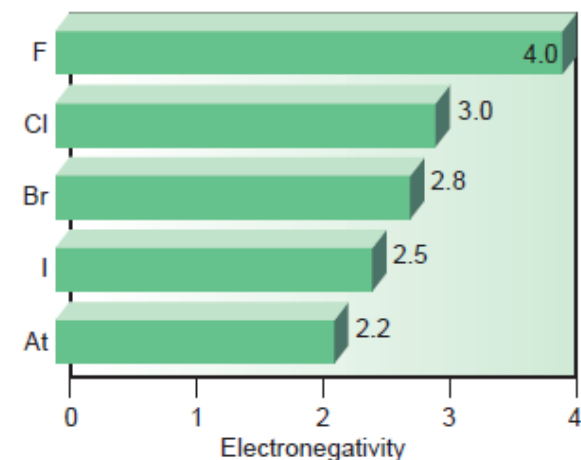
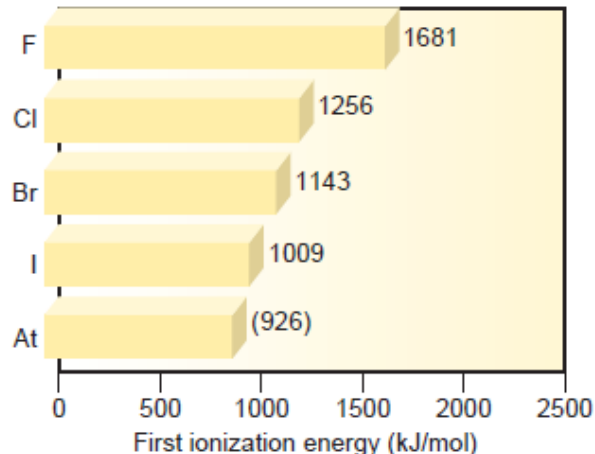
GROUP 7A(17)

Atomic Properties

Group electron configuration is ns^2np^5 ; elements lack one electron to complete their outer level. The -1 oxidation state is the most common for all members. Except for F, the halogens exhibit all odd-numbered states (+7 through -1).

Down the group, atomic and ionic size increase steadily, as IE and EN decrease.

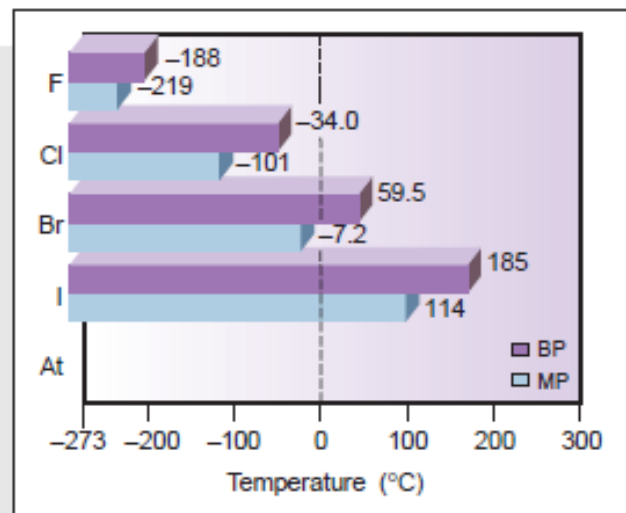
Atomic radius (pm)		Ionic radius (pm)
F 72		F ⁻ 133
Cl 100		Cl ⁻ 181
Br 114		Br ⁻ 196
I 133		I ⁻ 220
At (140)		no data



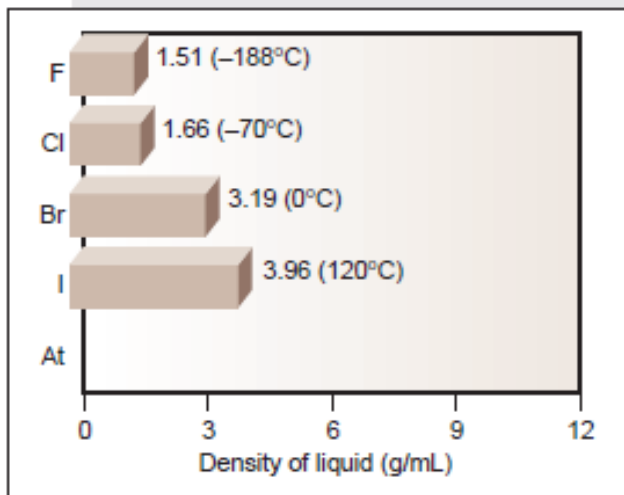
9 F 19.00 $2s^2 2p^5$ (-1)	Photograph not available
17 Cl 35.45 $3s^2 3p^5$ (-1, +7, +5, +3, +1)	
35 Br 79.90 $4s^2 4p^5$ (-1, +7, +5, +3, +1)	
53 I 126.9 $5s^2 5p^5$ (-1, +7, +5, +3, +1)	
85 At (210) $6s^2 6p^5$ (-1)	Extremely rare, no sample available

Physical Properties

Down the group, melting and boiling points increase smoothly as a result of stronger dispersion forces between heavier molecules.



The densities of the elements as liquids (at given T) increase steadily with molar mass.



hypohalites (XO^-)

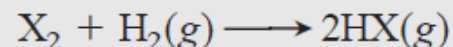
halites (XO^{2-})

halates (XO^{3-})

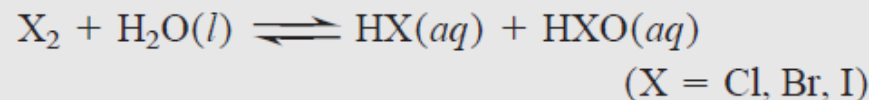
Important Reactions

Oxidizing strength and aqueous redox chemistry of the halogens are shown in reactions 1 and 2, and industrial processes involving fluorine in reactions 3 and 4.

1. The halogens (X_2) oxidize many metals and nonmetals. The reaction with hydrogen, although not used commercially for HX production (except for high-purity HCl), is characteristic of these strong oxidizing agents:



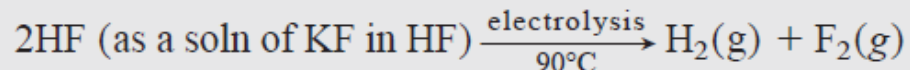
2. The halogens disproportionate in water:



In aqueous base, the reaction goes to completion to form hypohalites (*see text*) and, at higher temperatures, halates; for example:

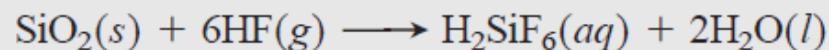


3. F_2 is produced electrolytically at moderate temperature:



A major use of F_2 is in the preparation of UF_6 for nuclear fuel.

4. Glass (amorphous silica) is etched with HF:



Important Compounds

1. Fluorspar (fluorite), CaF_2 . Widely distributed mineral used as a flux in steelmaking and in the production of HF (*see photo*).

2. Hydrogen fluoride, HF . Colorless, extremely toxic gas used to make F_2 , organic fluorine compounds, and polymers. Also used in aluminum manufacture and in glass etching (*margin note, p. 614*).



Fluorite

3. Hydrogen chloride, HCl . Extremely water-soluble gas that forms hydrochloric acid, which occurs naturally in the stomach fluids of mammals (humans produce 1.5 L of 0.1 M HCl daily) and in volcanic gases (from reaction of H_2O and sea salt). Made by reaction of NaCl and H_2SO_4 and as a byproduct of plastics (PVC) production. Used in the “pickling” of steel (removal of adhering oxides) and in the production of syrups, rayon, and plastic.

4. Sodium hypochlorite, NaClO , and calcium hypochlorite, $\text{Ca}(\text{ClO})_2$. Oxidizing agents used to bleach wood pulp and textiles and to disinfect swimming pools, foods, and sewage (also used to disinfect the Apollo 11 on return from the Moon). Household bleach is 5.25% NaClO by mass in water.

5. Ammonium perchlorate, NH_4ClO_4 . Strong oxidizing agent used in the space shuttle program (*see photo*).

6. Potassium iodide, KI . Most common soluble iodide. Added to table salt to prevent thyroid disease (goiter). Used in chemical analysis because it is easily oxidized to I_2 , which provides a colored end point.

7. Polychlorinated biphenyls, PCBs. Mixture of chlorinated organic compounds used as nonflammable insulating liquids in electrical transformers. Production discontinued because of persistence in the environment, where they become concentrated in the tissues of fish, birds, and mammals, and cause reproductive disturbances and possibly cancer.



Space shuttle

I. Electronic Configurations.

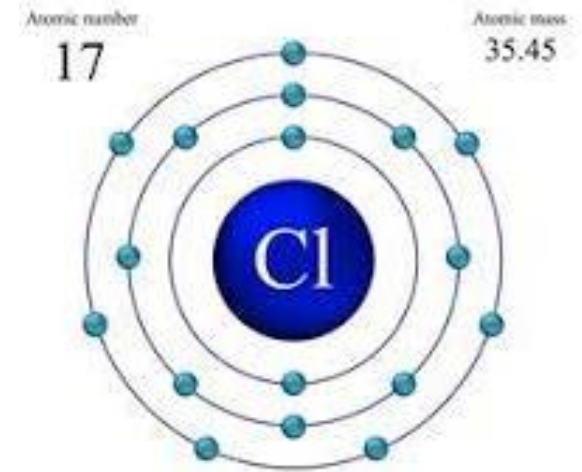
F (Z = 9) - (2 s²p⁵)

Cl (Z = 17) - (3 s²p⁵)

Br (Z = 35) - (4 s²p⁵)

I (Z = 53) - (5 s²p⁵)

At (Z = 85) - (6 s²p⁵)



- All the halogens have seven electrons (ns²p⁵) in their outer most shell.
- Two electrons are in ns-orbital and the remaining five electrons are in np-orbitals and are distributed as np²x, np²y, np¹z in accordance with Hund's rule.
- The shell previous to the outermost shell contains in F the group s² (saturated), in Cl s²p⁶ (saturated), in Br s²p⁶d¹⁰ (saturated) while in I and At s²p⁶d¹⁰ (unsaturated).
- This shows why F differs from Cl and these two from the remaining elements. Since the additional electron in these elements enters the p orbital of the outermost shell, these are called p-block elements of the long form of the periodic table.

Similarities in Physical Properties

1. Occurrence.

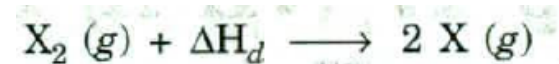
These elements do not occur in the free state. But in the combined state they occur fairly abundantly in nature.

2. Diatomic Nature.

- Exist as covalent diatomic molecules (X_2 molecule) in the gaseous, liquid and crystalline states.
- These X_2 molecules are held together by weak Van der Waals forces which explain the volatile nature of these elements.
- At high temperatures, X_2 molecules dissociate into X atoms.
- The stability of diatomic molecules towards heat decreases from F_2 to I_2

3. Heat of Dissociation of X_2 Molecules.

- Heat of dissociation (H_d) of X_2 molecule is the amount of heat required to dissociate X_2 molecule in the gaseous state into X gaseous atoms or to break X—X bond in X_2 molecule.



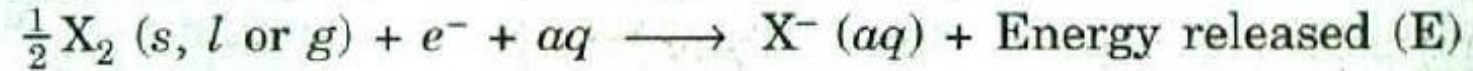
Obviously, ΔH_d is equal to X---X bond energy. Excepting for F_2 these energies decrease from Cl_2 to I_2 .

-The lower value of heat of dissociation of F_2 is due to the reasons;

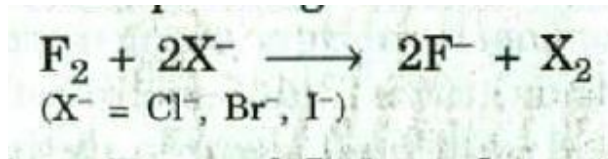
(i) Due to very small F—F bond length, repulsion between the non-bonding electrons is very large as compared to other X—X bond lengths (X = Cl, Br, I). This makes the F atoms in F_2 molecule repel each other and helps the dissociation of F_2 molecule into F atoms.

(ii) X—X bond in Cl_2 , Br_2 and I_2 molecules is stronger than F—F bond in F_2 molecule. This is due to the possibility of the existence of multiple bonding in X—X bond involving d-orbitals.

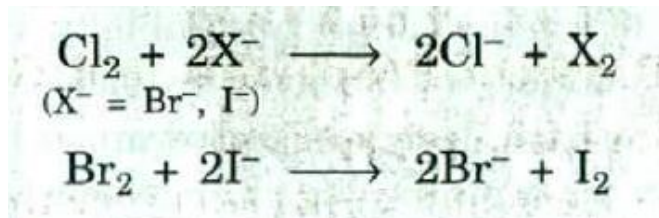
4. Oxidizing Property. A substance that has a tendency to accept one or more electrons is said to show oxidizing property. The halogens, due to high electron affinity values, have a great tendency to accept an electron and hence act as strong oxidizing agent.



- Since the values of E are decreasing from F_2 to I_2 the oxidizing power of halogens is also decreasing in the same direction, i.e. the oxidizing power of halogens is in the order: F_2 (strongest oxidizing agent) $>$ $Cl_2 > Br_2 > I_2$ (weakest oxidizing agent).
- Since F_2 is the strongest oxidizing agent in the series, it will oxidize other halide ions to halogens in solution or when dry, i.e. F_2 displaces other halogens from their corresponding halides.



- Similarly Cl_2 will displace Br^- and I^- ions from their solutions and Br_2 will displace I^- ions from their solutions.



5. Formation of Clathrates.

Water solutions of the halogens on cooling give the crystals of their clathrates of $X_2 \cdot xH_2O$ type.

Examples are : $Cl_2 \cdot 7.3 H_2O$, $Br_2 \cdot 10H_2O$, $I_2 \cdot xH_2O$ etc.

6. Electronegativity.

Halogens have large values of electronegativity. These values decrease from F to I in the group. Large electronegativity values of halogen atoms indicate that X atoms have a strong tendency to form X^- ions.

7. Solubility.

- The halogens are slightly soluble in water and their solubility decreases from Cl to I.
- F cannot be dissolved in H_2O , since it decomposes H_2O liberating O_2 and some O_3 .
- The halogens dissolve much better than H_2O in many organic solvents like carbon disulphide, chloroform, ether etc.
- In organic solvents, iodine gives violet colour, bromine yields brown colour and chlorine produces yellow colour.
- It is assumed that in organic solvents halogens exist as free molecule as in the gas phase. In nucleophilic (electron donating) polar solvent like alcohols, ketones the halogens give coloured solutions whose colours are due to the formation of solvent -- halogen type complexes. These complexes are charge-transfer compounds.

Similarities in Chemical Properties

1. Reactivity.

- All the halogens are the most reactive elements as a family.

The main reasons for their extreme reactivity are :

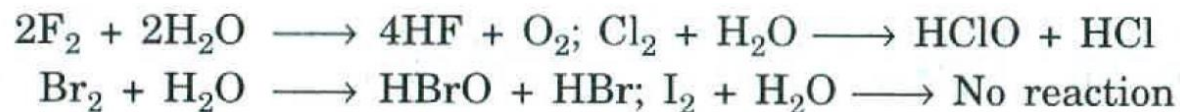
(a) Low dissociation energies of X_2 molecules.

For example: $O_2 = 495.08$, $F_2 = 157.77$, $Cl_2 = 237.38$, $Br_2 = 189.99$ in kJ/mol.

(b) High electron affinity values of halogen atoms - halogens have a strong tendency to add on electron to form X ion.

The reactivity of halogen atoms decreases down the group from F to I.

- For example:
- fluorine decomposes H_2O vigorously even in dark,
 - chlorine decomposes H_2O in sunlight
 - bromine also decomposes H_2O in sunlight but very slowly
 - iodine does not decompose water at all



2. Formation of Hydracids.

- All the elements of this group combine with hydrogen and form hydrogen halides (HX) which are called hydracids or halogen acids.
- The affinity of these elements for hydrogen decreases from fluorine to iodine.
 - F₂ combines with H₂ explosively and even in the dark
 - Cl₂ reacts with H₂ in diffused sunlight,
 - Br₂ reacts with H₂ only on heating above 500 °C
 - I₂ reacts with H₂ only on heating and in presence of Pt as a catalyst.
- Methods of formation of Hydracids are such as (i) direct combination of the elements (ii) heating the metal halides with conc. H₂SO₄ (iii) hydrolysis of non-metallic halides.
- *Some important properties of hydracids are given below.*

(i) Acidic properties.

All the halogen acids in the gaseous state are essentially covalent but in the aqueous solution they ionize to give the solvated proton (H₃O⁺) and hence act as acids.



HF ionizes only slightly while HCl, HBr and HI ionize completely. Hence HF is the weakest acid and the strength of these acids increases from HF to HI,

i.e. HF (weakest acid) < HCl < HBr < HI (strongest acid).

The weakest acidic nature of HF is due to the fact that the dissociation energy of H—F bond in HF molecule is the highest (= 564.97 KJ.mol⁻¹) and hence this molecule has least tendency to split up into H⁺ and F⁻ ions in aqueous solution.

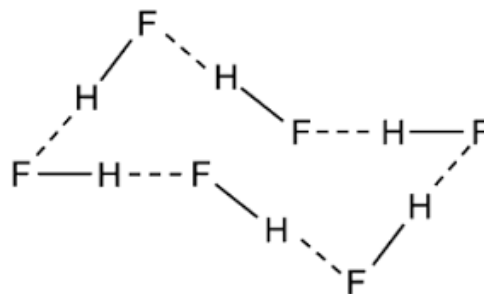
(ii) Covalent character.

- The bond between hydrogen and the halogen in these hydrides is largely covalent.
- The bonds do, however, have small amount of polar character which decreases with the decrease of electronegativity of the halogen atom. Thus, the polar character of H—X bond in HX molecules decreases in the order:



(iii) Physical properties.

- Anhydrous HF is a liquid at ordinary temperature whereas other HX (X = Cl, Br, I) are colourless gases.
- Excepting HF, the melting and boiling points of these hydrides increase with the increase of the atomic weight of the halogen atom.
- The abnormally high melting point of HF is explained by saying that, due to H-bonding, HF molecule gets associated to form (HF)₆ which is an aggregated molecule.



- The values of heat of formation increase with the increase of atomic weight of halogen atom. Thus the quantity of energy released during the formation of HX molecules decreases from HF to HI.

(iv) **Stability and reducing property.**

- As we go from HF to HI, the stability of these acids decreases from HF to HI. (HF > HCl > HBr > HI).
- Thus HF and HCl are stable upto 1200°C while HBr dissociates slightly (I) and HI dissociates at 440°C to 20%. The decrease in the stability of HX molecules from HF to HI implies that the reducing properties of these acids increase from HF to HI,

i.e. HF < HCl < HBr < HI.

- The above order shows that HF has no appreciable reducing properties while HI is the strongest reducing agent.
- Even dilute solution of HI, on exposure to the air, slowly acquires a brown colour due to the liberation of I₂.



3. Formation of Binary Halogen-oxygen Compounds.

- Halogens do not combine directly with O₂. However, binary halogen-oxygen compounds can be prepared by indirect methods.
- Eg: OF₂, O₂F₂, Cl₂O, ClO₂, Br₂O, BrO₂, I₂O₄.
- F being more electronegative than oxygen, OF₂ and O₂F₂ are called fluorides of oxygen and not oxides of fluorine.

4. Formation of Oxy-acids.

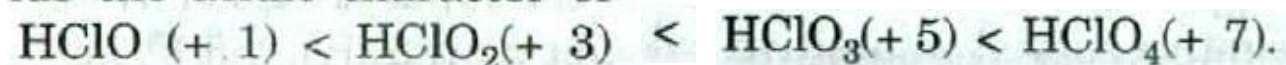
- Excepting F, all other halogens form oxy-acids.
- These are mainly of four types

- HXO (e.g. HClO, HBrO, HIO)
- HXO₂ (e.g. HClO₂)
- HXO₃ (e.g. HClO₄, HBrO₄, HIO₄)
- HXO₄ (e.g. HClO₄, HBrO₄, HIO₄)

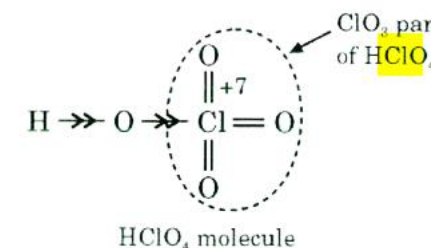
Some of their properties are given below

(i) Acidic character.

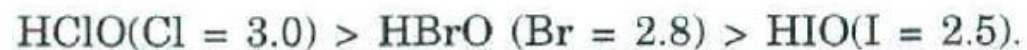
(a) In the series of oxy-acids of the same halogen in different oxidation number the acidic character of such acids increases with the increase of the oxidation number of the halogen. Thus the acidic character of HClO, HClO₂, HClO₃, HClO₄ acids is in the order.



The above order of acidic character of the acids is explained by finding out the relative order of the basicity of the conjugate bases (viz. ClO, ClO₂, ClO₃ and ClO₄) of these acids.



(b) The acidic character of the oxy-acids of different halogens in the same oxidation state decreases, in general, with decrease of the electronegativity of the halogen atom. Thus the acidic character of HClO, HBrO and HIO acids is in the following order.



Electronegativity of halogen atoms is given in bracket.

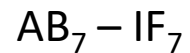
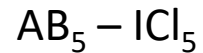
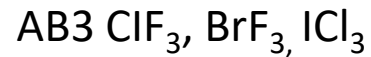
- As the atomic number of the halogen atom increases, its electronegativity decreases and size increases; As a result of this, the tendency of these acids to lose a proton to water decreases. This makes the acid a weaker acid.

(ii) Oxidizing power.

- As the oxidation number of the halogen atom increases, X—O bond becomes more covalent, resulting in a decrease of the oxidizing power of the acid.
- For example under usual laboratory conditions, the oxidizing action of HClO, HClO₂, HClO₃, and HClO₄ acids is in the order HClO(+ 1) > HClO₂(+ 3) > HClO₃(+ 5) > HClO₄(+ 7).

5. Formation of Interhalogen Compounds.

- Two different halogen atoms, due to the difference in their electronegativity values, combine with one another to form a number of covalent compounds which are called interhalogen compounds.



Anomalous Behaviour of Fluorine

- Fluorine differs from the other members of its own subgroup due to
 - (a) its small size
 - (b) its high electronegativity
 - (c) non-availability of d-orbitals in its valence shell.

The main differences are :

- (i) Due to its maximum electronegativity, it shows only a negative oxidation state of -1. It does not show any positive oxidation state. The other members show negative as well as positive oxidation states of + 1, + 3, + 5 and + 7.
- (ii) Fluorine is the most reactive element among halogens. This is due to the minimum value of F—F bond dissociation energy.
- (iii) On account of high electronegativity of fluorine atom hydrogen bonding occurs in HF molecule while it is absent in HCl, HBr and HI molecules. The presence of hydrogen bonding in HF explains why
 - (a) HCl, HBr and HI are gaseous while HF is a liquid with an abnormally high boiling point
 - (b) HF forms a number of compounds containing HF_2^- ion are not given by other HX molecules.
- (iv) HF ionizes only slightly in aqueous solution and is, therefore a weak acid. HCl, HBr and HI, on the other hand, ionize almost completely and are thus strong acids.

- (v) Fluoride of silver is soluble while other halides of silver are insoluble. Similarly fluoride of calcium is insoluble and other halides of calcium are soluble.
- (vi) Fluorides are ionic while other halides have frequently molecular lattices.
- (vii) F ion has a great tendency to give complex ions like $[\text{AlF}_6]^{3-}$, $[\text{FeF}_6]^{3-}$ etc. The remaining halide ions show this tendency to a much lesser extent.
- (viii) Fluorine, being the most electronegative, gives SF_6 while other members do not form hexahalides with sulphur.
- (x) Because of the absence of d-orbitals in its valence shell, fluorine does not combine with F ions to give polyhalide ion, F_3^- while other members of the group give such ions (e.g. Cl_3^- , Br_3^- , I_3^- , I_5^- etc.) because they contain d-orbitals.

Comparison of Fluorine and Oxygen.

Fluorine resembles oxygen on following factors :

- (a) The atomic and ionic radii of both the elements are very similar
- (b) Both are highly electronegative.

- **Similarities.**

- (i) Both of them exist in gaseous state at ordinary temperature
- (ii) Both react with metals (e.g. Na) and non- metals (e.g., H, C, Si, S etc.) to give fluorides and oxides which resemble each other in solubility, volatility, fusibility etc., e.g.

HF and H₂O resemble each other in the following respects

- (a) Both possess associated molecules, (H₂O) and (HF) respectively due to hydrogen bonding
- (b) The boiling points of both are abnormally high
- (c) HF like H₂O is a weak acid and both are ionizing solvents.
- (d) Heats of formation of HF and H₂O are close to each other.

(iii) Certain elements like sulphur exhibit higher valencies in both fluorides and oxides, e.g. the valency of sulphur in SF₆, and SO₃ is 6.

Dissimilarities.

However these elements differ from each other in the following respects :

- (i) F is always monovalent while oxygen is bivalent
- (ii) F is more reactive than oxygen.